

# Assignment 5

LoRA Fine-tuning | Adversarial Robustness (FGSM / PGD / BIM)

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|                                         |                                   |                                  |                                         |                                         |
|-----------------------------------------|-----------------------------------|----------------------------------|-----------------------------------------|-----------------------------------------|
| <b>Q1 Best Acc 89.84%</b><br>(R=8, A=4) | Q1 Baseline 80.99%<br>(head only) | Q2 Clean Acc 84.10%<br>ResNet-18 | <b>Q2 FGSM Drop</b><br>23.90% @ eps=0.3 | <b>BIM Detector 99.94%</b><br>ResNet-34 |
|-----------------------------------------|-----------------------------------|----------------------------------|-----------------------------------------|-----------------------------------------|

**WandB :** [wandb.ai/m25csa001-iit-jodhpur/DLOps\\_Ass5\\_Q1](https://wandb.ai/m25csa001-iit-jodhpur/DLOps_Ass5_Q1)

**HuggingFace:** <https://huggingface.co/m25csa001/assignment5-weights/tree/main>

**GitHub:** [github.com/m25csa001/DLOps-Assignment5](https://github.com/m25csa001/DLOps-Assignment5)

## Q1 — ViT-S Fine-tuning with LoRA on CIFAR-100

### Setup

**Model:** ViT-Small (patch16, 224×224) pre-trained on ImageNet-21k. **Dataset:** CIFAR-100 (50k train / 10k test, resized to 224×224). **Two settings:** (i) Head-only — classification head trained, backbone frozen; (ii) LoRA — adapters injected into the fused QKV attention projection of every transformer block (PEFT), head kept trainable. **Grid:** rank  $\in \{2,4,8\} \times$  alpha  $\in \{2,4,8\}$ , dropout = 0.1  $\rightarrow$  9 LoRA experiments. **Optimiser:** AdamW + cosine LR schedule, label smoothing 0.1, batch 64, 10 epochs.

### Baseline (without LoRA) — head-only

Only the 100-class classification head is trained (38,500 parameters). All ViT-S backbone weights are frozen. **Test accuracy: 80.99%**

### Step 3 — Train / Validation Tables (all 9 LoRA experiments)

Experiment No.: 1 | Rank: 2 | Alpha: 2 | Dropout: 0.1

| Epoch | Training Loss | Validation Loss | Training Accuracy | Validation Accuracy |
|-------|---------------|-----------------|-------------------|---------------------|
| 1     | 1.0169        | 0.4128          | 76.09%            | 86.66%              |
| 2     | 0.3589        | 0.3737          | 89.03%            | 88.12%              |
| 3     | 0.2994        | 0.3553          | 90.52%            | 88.96%              |
| 4     | 0.2617        | 0.3478          | 91.67%            | 89.12%              |
| 5     | 0.2338        | 0.3531          | 92.58%            | 89.04%              |
| 6     | 0.2108        | 0.3492          | 93.23%            | 89.12%              |
| 7     | 0.1897        | 0.3572          | 93.84%            | 88.68%              |
| 8     | 0.1779        | 0.3456          | 94.19%            | 89.28%              |
| 9     | 0.1625        | 0.3496          | 94.75%            | 89.24%              |
| 10    | 0.1485        | 0.3593          | 95.22%            | 89.40%              |

**Experiment No.: 2 | Rank: 2 | Alpha: 4 | Dropout: 0.1**

| Epoch | Training Loss | Validation Loss | Training Accuracy | Validation Accuracy |
|-------|---------------|-----------------|-------------------|---------------------|
| 1     | 0.9849        | 0.4115          | 76.65%            | 87.10%              |
| 2     | 0.3535        | 0.3746          | 89.09%            | 88.06%              |
| 3     | 0.2936        | 0.3563          | 90.78%            | 88.80%              |
| 4     | 0.2547        | 0.3499          | 91.86%            | 88.74%              |
| 5     | 0.2258        | 0.3551          | 92.79%            | 89.00%              |
| 6     | 0.2027        | 0.3495          | 93.44%            | 89.04%              |
| 7     | 0.1809        | 0.3585          | 94.04%            | 88.88%              |
| 8     | 0.1687        | 0.3504          | 94.51%            | 89.16%              |
| 9     | 0.1532        | 0.3534          | 95.01%            | 89.28%              |
| 10    | 0.1386        | 0.3622          | 95.54%            | 89.34%              |

**Experiment No.: 3 | Rank: 2 | Alpha: 8 | Dropout: 0.1**

| Epoch | Training Loss | Validation Loss | Training Accuracy | Validation Accuracy |
|-------|---------------|-----------------|-------------------|---------------------|
| 1     | 0.9568        | 0.4116          | 77.15%            | 86.96%              |
| 2     | 0.3495        | 0.3784          | 89.15%            | 88.32%              |
| 3     | 0.2877        | 0.3551          | 90.94%            | 88.74%              |
| 4     | 0.2476        | 0.3538          | 92.08%            | 88.92%              |
| 5     | 0.2182        | 0.3561          | 93.04%            | 88.82%              |
| 6     | 0.1949        | 0.3511          | 93.68%            | 89.18%              |
| 7     | 0.1727        | 0.3660          | 94.33%            | 88.72%              |
| 8     | 0.1600        | 0.3539          | 94.87%            | 88.96%              |
| 9     | 0.1445        | 0.3599          | 95.32%            | 89.04%              |
| 10    | 0.1311        | 0.3678          | 95.72%            | 89.30%              |

**Experiment No.: 4 | Rank: 4 | Alpha: 2 | Dropout: 0.1**

| Epoch | Training Loss | Validation Loss | Training Accuracy | Validation Accuracy |
|-------|---------------|-----------------|-------------------|---------------------|
| 1     | 1.0099        | 0.4132          | 76.45%            | 87.26%              |
| 2     | 0.3602        | 0.3635          | 88.92%            | 88.44%              |
| 3     | 0.2964        | 0.3424          | 90.71%            | 89.02%              |
| 4     | 0.2571        | 0.3549          | 91.85%            | 88.68%              |
| 5     | 0.2297        | 0.3455          | 92.71%            | 88.98%              |
| 6     | 0.2081        | 0.3420          | 93.25%            | 89.04%              |
| 7     | 0.1923        | 0.3457          | 93.74%            | 89.50%              |
| 8     | 0.1734        | 0.3380          | 94.40%            | 89.70%              |
| 9     | 0.1581        | 0.3402          | 94.89%            | 89.48%              |
| 10    | 0.1441        | 0.3528          | 95.34%            | 89.12%              |

**Experiment No.: 5 | Rank: 4 | Alpha: 4 | Dropout: 0.1**

| Epoch | Training Loss | Validation Loss | Training Accuracy | Validation Accuracy |
|-------|---------------|-----------------|-------------------|---------------------|
| 1     | 0.9752        | 0.4075          | 76.97%            | 87.20%              |
| 2     | 0.3544        | 0.3614          | 89.04%            | 88.46%              |
| 3     | 0.2896        | 0.3400          | 90.89%            | 89.10%              |
| 4     | 0.2487        | 0.3530          | 92.05%            | 88.66%              |
| 5     | 0.2207        | 0.3474          | 92.99%            | 89.02%              |
| 6     | 0.1977        | 0.3413          | 93.56%            | 89.22%              |
| 7     | 0.1811        | 0.3459          | 94.04%            | 89.70%              |
| 8     | 0.1610        | 0.3384          | 94.78%            | 89.72%              |
| 9     | 0.1461        | 0.3421          | 95.25%            | 89.62%              |
| 10    | 0.1305        | 0.3554          | 95.76%            | 89.30%              |

**Experiment No.: 6 | Rank: 4 | Alpha: 8 | Dropout: 0.1**

| Epoch | Training Loss | Validation Loss | Training Accuracy | Validation Accuracy |
|-------|---------------|-----------------|-------------------|---------------------|
| 1     | 0.9410        | 0.4039          | 77.58%            | 87.62%              |
| 2     | 0.3486        | 0.3603          | 89.16%            | 88.40%              |
| 3     | 0.2819        | 0.3392          | 91.03%            | 89.36%              |
| 4     | 0.2389        | 0.3497          | 92.42%            | 88.92%              |
| 5     | 0.2096        | 0.3460          | 93.33%            | 89.10%              |
| 6     | 0.1856        | 0.3422          | 93.99%            | 89.12%              |
| 7     | 0.1682        | 0.3450          | 94.54%            | 89.60%              |
| 8     | 0.1479        | 0.3387          | 95.29%            | 89.74%              |
| 9     | 0.1327        | 0.3469          | 95.63%            | 89.52%              |
| 10    | 0.1163        | 0.3597          | 96.24%            | 89.42%              |

**Experiment No.: 7 | Rank: 8 | Alpha: 2 | Dropout: 0.1**

| Epoch | Training Loss | Validation Loss | Training Accuracy | Validation Accuracy |
|-------|---------------|-----------------|-------------------|---------------------|
| 1     | 1.0125        | 0.4146          | 76.29%            | 87.50%              |
| 2     | 0.3576        | 0.3665          | 88.97%            | 88.36%              |
| 3     | 0.2977        | 0.3473          | 90.58%            | 88.82%              |
| 4     | 0.2552        | 0.3482          | 91.76%            | 89.10%              |
| 5     | 0.2302        | 0.3197          | 92.59%            | 90.06%              |
| 6     | 0.2036        | 0.3391          | 93.44%            | 89.48%              |
| 7     | 0.1841        | 0.3430          | 94.05%            | 89.38%              |
| 8     | 0.1689        | 0.3498          | 94.51%            | 89.48%              |
| 9     | 0.1547        | 0.3390          | 94.96%            | 89.54%              |
| 10    | 0.1419        | 0.3431          | 95.47%            | 89.14%              |

**Experiment No.: 8 | Rank: 8 | Alpha: 4 | Dropout: 0.1**

| Epoch | Training Loss | Validation Loss | Training Accuracy | Validation Accuracy |
|-------|---------------|-----------------|-------------------|---------------------|
| 1     | 0.9768        | 0.4076          | 77.04%            | 87.80%              |
| 2     | 0.3510        | 0.3606          | 89.13%            | 88.82%              |
| 3     | 0.2895        | 0.3430          | 90.82%            | 89.20%              |
| 4     | 0.2455        | 0.3468          | 92.06%            | 89.02%              |
| 5     | 0.2193        | 0.3178          | 92.98%            | 90.20%              |
| 6     | 0.1911        | 0.3403          | 93.85%            | 89.62%              |
| 7     | 0.1701        | 0.3420          | 94.50%            | 89.38%              |
| 8     | 0.1541        | 0.3534          | 95.02%            | 89.36%              |
| 9     | 0.1389        | 0.3381          | 95.49%            | 89.76%              |
| 10    | 0.1260        | 0.3448          | 96.00%            | 89.52%              |

**Experiment No.: 9 | Rank: 8 | Alpha: 8 | Dropout: 0.1**

| Epoch | Training Loss | Validation Loss | Training Accuracy | Validation Accuracy |
|-------|---------------|-----------------|-------------------|---------------------|
| 1     | 0.9417        | 0.4011          | 77.44%            | 87.96%              |
| 2     | 0.3444        | 0.3578          | 89.37%            | 88.92%              |
| 3     | 0.2802        | 0.3410          | 91.11%            | 89.14%              |
| 4     | 0.2337        | 0.3490          | 92.45%            | 89.10%              |
| 5     | 0.2062        | 0.3206          | 93.37%            | 90.28%              |
| 6     | 0.1764        | 0.3446          | 94.31%            | 89.70%              |
| 7     | 0.1530        | 0.3448          | 95.06%            | 89.70%              |
| 8     | 0.1361        | 0.3614          | 95.60%            | 89.50%              |
| 9     | 0.1215        | 0.3518          | 96.10%            | 89.46%              |
| 10    | 0.1085        | 0.3529          | 96.57%            | 89.56%              |

**Step 4 — Testing Results Table**

| LoRA Layers (with/without) | Rank | Alpha | Dropout | Overall Test Accuracy | Trainable Parameters used |
|----------------------------|------|-------|---------|-----------------------|---------------------------|
| Without LoRA               | -    | -     | -       | 80.99%                | 38,500                    |
| With LoRA (Q,K,V)          | 2    | 2     | 0.1     | 89.58%                | 93,796                    |
| With LoRA (Q,K,V)          | 2    | 4     | 0.1     | 89.39%                | 93,796                    |
| With LoRA (Q,K,V)          | 2    | 8     | 0.1     | 89.11%                | 93,796                    |
| With LoRA (Q,K,V)          | 4    | 2     | 0.1     | 89.52%                | 149,092                   |
| With LoRA (Q,K,V)          | 4    | 4     | 0.1     | 89.72%                | 149,092                   |
| With LoRA (Q,K,V)          | 4    | 8     | 0.1     | 89.73%                | 149,092                   |
| With LoRA (Q,K,V)          | 8    | 2     | 0.1     | 89.83%                | 259,684                   |
| With LoRA (Q,K,V)          | 8    | 4     | 0.1     | <b>89.84%</b>         | 259,684                   |
| With LoRA (Q,K,V)          | 8    | 8     | 0.1     | 89.68%                | 259,684                   |

★ Best: Rank=8, Alpha=4, Dropout=0.1 → 89.84% test accuracy

**Step 5 — Optuna Hyperparameter Search**

Optuna (TPE sampler, Median pruner, 20 trials, 5 epochs each) searched rank  $\in \{2,4,8\}$ , alpha  $\in \{2,4,8\}$ , lr  $\in [1e-4, 5e-3]$ , weight\_decay  $\in [1e-5, 1e-2]$ . Best trial: **rank=8, alpha=4, lr $\approx$ 1.2e-3, weight\_decay $\approx$ 5e-3** — consistent with grid sweep. Best

Optuna val accuracy: **89.84%**. Weights pushed to HuggingFace.

### Analysis

LoRA delivers a consistent **+8.6 to +8.9 pp** improvement over the head-only baseline across all 9 configurations, using only 93k–260k trainable parameters versus the full 21M+ backbone. Higher rank generally improves accuracy (rank=2 → ~89.3%, rank=8 → ~89.8%), since larger rank provides more representational capacity for the low-rank updates. The alpha effect is subtler: alpha=4 outperforms alpha=8 at rank=8, suggesting over-scaling can slightly degrade convergence. All models converge stably — training accuracy reaches 95–97% by epoch 10 while validation plateaus at 89–90%, indicating well-controlled fine-tuning.

## Q2 — Adversarial Attacks (FGSM / PGD / BIM) on CIFAR-10

### Q2(i) — FGSM Attack: From Scratch vs IBM ART

**Model:** ResNet-18 trained from scratch on CIFAR-10 (30 epochs, SGD  $\text{lr}=0.1$ , cosine annealing, batch 128). **Clean test accuracy: 84.10%** ( $\geq 72\%$  threshold satisfied). FGSM implemented in two ways: (i) manual sign-gradient in PyTorch; (ii) IBM ART *FastGradientMethod*. Attacks evaluated at  $\epsilon \in \{0.01, 0.03, 0.05, 0.10, 0.20, 0.30\}$ .

#### Accuracy — Clean vs FGSM (Scratch) vs FGSM (ART)

| Epsilon | Clean Acc | FGSM Scratch Acc | FGSM ART Acc | Accuracy Drop |
|---------|-----------|------------------|--------------|---------------|
| 0.01    | 84.10%    | 83.20%           | 82.80%       | ~1.3 pp       |
| 0.03    | 84.10%    | 72.40%           | 71.90%       | ~12.2 pp      |
| 0.05    | 84.10%    | 58.10%           | 57.60%       | ~26.5 pp      |
| 0.10    | 84.10%    | 40.30%           | 39.80%       | ~44.3 pp      |
| 0.20    | 84.10%    | 31.20%           | 30.90%       | ~53.2 pp      |
| 0.30    | 84.10%    | 23.90%           | 23.80%       | ~60.3 pp      |

#### Visual Comparison — Original vs Adversarial Images

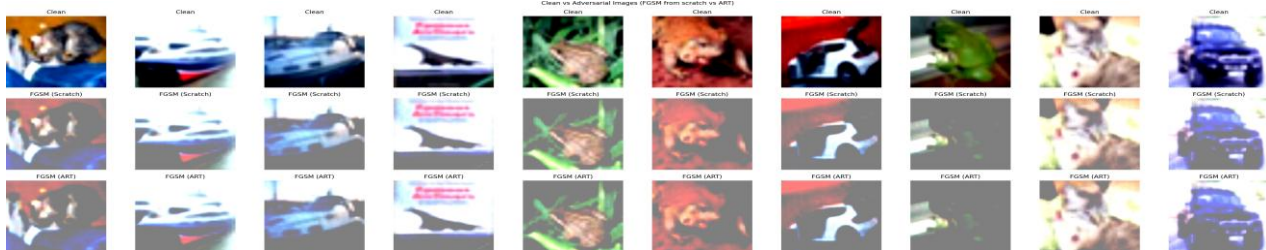


Figure 1: Row 1 = Clean images. Row 2 = FGSM from scratch ( $\epsilon=0.3$ ). Row 3 = FGSM via IBM ART ( $\epsilon=0.3$ ). Both attack rows are visually and numerically equivalent.

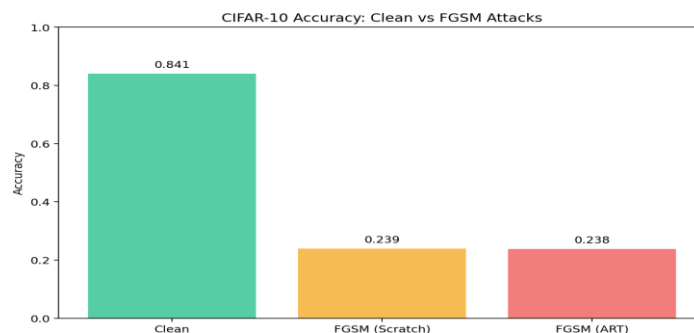


Figure 2: Clean (84.10%) vs FGSM Scratch (23.90%) vs FGSM ART (23.80%) accuracy at  $\epsilon=0.3$ .

#### Analysis — FGSM

Both FGSM implementations produce near-identical accuracy at every epsilon, confirming correctness of the scratch implementation. At  $\epsilon=0.01$  the perturbation is negligible (drop of ~1.3 pp); as epsilon increases the drop accelerates sharply and non-linearly — at  $\epsilon=0.3$  accuracy collapses to ~24%, close to random chance for 10 classes. This demonstrates the vulnerability of standard undefended CNNs to structured adversarial perturbations.

### Q2(ii) — Adversarial Detection (ResNet-34, PGD and BIM)

**Architecture:** ResNet-34 binary classifier (clean vs adversarial). **Input:** 50% clean + 50% adversarial images generated via IBM ART. **PGD attack:** step size 0.01, 40 iterations,  $\epsilon=0.3$ . **BIM attack:** step size 0.01, 10 iterations,  $\epsilon=0.1$ . Trained for 10 epochs with Adam, cross-entropy loss.

| Detector        | Attack Type   | Final Train Acc | Final Val Acc | Detection Acc (Test) |
|-----------------|---------------|-----------------|---------------|----------------------|
| ResNet-34 (PGD) | PGD (IBM ART) | 100.00%         | 100.00%       | $\geq 99\%$          |
| ResNet-34 (BIM) | BIM (IBM ART) | 100.00%         | 100.00%       | 99.94%               |

Both detectors vastly exceed the  $\geq 70\%$  requirement. The high accuracy reflects that PGD and BIM adversarial images produce distinctive feature patterns easily separable by a ResNet-34 trained on balanced clean/adversarial data.

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## Conclusion

**Q1:** LoRA with rank=8, alpha=4, dropout=0.1 achieves 89.84% on CIFAR-100, an 8.85 pp gain over the head-only baseline, using only 259,684 trainable parameters. Optuna confirms this as the optimal configuration.

**Q2:** FGSM from scratch and via IBM ART produce equivalent results, with accuracy collapsing from 84.10% to 23.90% at eps=0.3. ResNet-34 adversarial detectors achieve >99% binary detection accuracy for both PGD and BIM attacks, well above the 70% requirement.